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MachineLearning

EXPLORER

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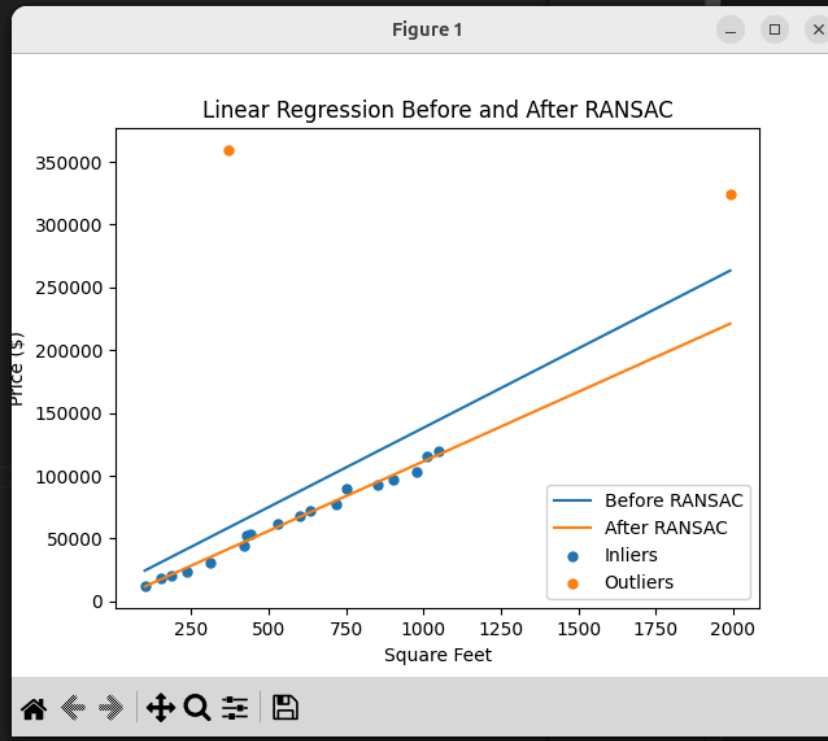
WilliamsBenjamin_Lab4/... | image.png .../Exercise2 | image.png .../Exercise3 | exercise4.py | image copy.png | exercise1.py .../WilliamsBenjamin_HW3/... | exercise2.py .../WilliamsBenjamin_HW3/...

HW > HW3 > WilliamsBenjamin_HW3 > Exercise1 > exercise1.py > ...

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression, RANSACRegressor

X = np.array([100, 150, 185, 235, 310, 370, 420, 430, 440, 530, 600, 634, 718, 750, 850, 903, 978, 1010, 1050, 1990], dtype=float).reshape(-1, 1)
y = np.array([12300, 18150, 20100, 23500, 31005, 359000, 44359, 52000, 53853, 61328, 68000, 72300, 77000, 89379, 93200, 97150, 102750, 115358, 119330, 323989], dtype=flo

lr = LinearRegression()
lr.fit(X, y)
m1 = lr.coef_[0]
b1 = lr.intercept_

ransac = RANSACRegressor(estimator=LinearRegression(), random_state=0)
ransac.fit(X, y)
inlier_mask = ransac.inlier_mask_
outlier_mask = ~inlier_mask

m2 = ransac.estimator_.coef_[0]
b2 = ransac.estimator_.intercept_

print("Before RANSAC: slope =", m1, ", y-intercept =", b1)
print("After  RANSAC: slope =", m2, ", y-intercept =", b2)

x_line = np.linspace(X.min(), X.max(), 200).reshape(-1, 1)
y_line_before = lr.predict(x_line)
y_line_after = ransac.predict(x_line)

plt.figure()
plt.plot(x_line, y_line_before, label="Before RANSAC")
plt.plot(x_line, y_line_after, label="After RANSAC")
plt.scatter(X[inlier_mask], y[inlier_mask], s=25, label="Inliers")
plt.scatter(X[outlier_mask], y[outlier_mask], s=25, label="Outliers")
plt.xlabel("Square Feet")
plt.ylabel("Price ($)")
plt.title("Linear Regression Before and After RANSAC")
plt.legend()
plt.show()
```



PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW3/WilliamsBenjamin_HW3/Exercis
e1/exercise1.py
  File "/home/ben/.local/lib/python3.14/site-packages/matplotlib/backends/_backend_tk.py", line 572, in start_main_loop
    first_manager.window.mainloop()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "/usr/local/lib/python3.14/tkinter/__init__.py", line 1613, in mainloop
    self.tk.mainloop(n)
    ~~~~~~~~~~~~~~~~^^^
KeyboardInterrupt

ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW3/WilliamsBenjamin_HW3/Exercis
e1/exercise1.py
Before RANSAC: slope = 126.412859778847 , y-intercept = 11727.454260912462
After  RANSAC: slope = 110.76198060341628 , y-intercept = 610.4963138353487
```

bash | bash Ex... | python3.14

OUTLINE | TIMELINE

main* | 0 0 | Ln 31, Col 52 | Spaces: 4 | UTF-8 | LF | Python

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OUTLINE

TIMELINE

image.png .../Exercise3 | exercise4.py | image copy.png | exercise1.py .../WilliamsBenjamin_HW3/... | image.png .../WilliamsBenjamin_HW3/... | exercise2.py .../WilliamsBenjamin_HW3/... | iris.data.csv

HW > HW3 > WilliamsBenjamin_HW3 > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier

df = pd.read_csv("iris.data.csv", header=None)
X = df.iloc[:, 0:4].astype(float).to_numpy()
y = df.iloc[:, 4].astype(str).to_numpy()

n = len(X)
kfolds = 5
fold_sizes = [30] * 5

indices = np.arange(n)
folds = []
start = 0
for fs in fold_sizes:
    folds.append(indices[start:start + fs])
    start += fs

accuracies = []
for i in range(kfolds):
    test_idx = folds[i]
    train_idx = np.hstack([folds[j] for j in range(kfolds) if j != i])

    X_train, y_train = X[train_idx], y[train_idx]
    X_test, y_test = X[test_idx], y[test_idx]

    knn = KNeighborsClassifier(n_neighbors=9)
    knn.fit(X_train, y_train)
    acc = knn.score(X_test, y_test)
    accuracies.append(acc)
    print(f"Fold {i+1} accuracy:", acc)

print("Average accuracy:", float(np.mean(accuracies)))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW3/WilliamsBenjamin_HW3/Exercise2$ ls
exercise2.py  iris.data.csv
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW3/WilliamsBenjamin_HW3/Exercise2$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearni
ng/HW/HW3/WilliamsBenjamin_HW3/Exercise2/exercise2.py
Fold 1 accuracy: 1.0
Fold 2 accuracy: 1.0
Fold 3 accuracy: 0.8666666666666667
Fold 4 accuracy: 0.9666666666666667
Fold 5 accuracy: 0.7666666666666667
Average accuracy: 0.9200000000000002
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW3/WilliamsBenjamin_HW3/Exercise2$
```

bash | bash Ex... | bash Exerci...

Do you want to install the recommende
extension from mechatroner for iris.da
Inst

CHAT

SESSIONS

Feature Exp
Completed i

main* 0 0 | Ln 34, Col 54 | Spaces: 4 | UTF-8 | LF | Python